TLS requires them to also send a *CertificateVerify* message in which they sign a transcript of all previous messages exchanged during the handshake. This proves that the client (or server) has control of the private key corresponding to their certificate and ensures that the signature is tightly bound to this specific handshake: there are unique values exchanged in the handshake, preventing the signature being reused for any other TLS session. The session keys used for authenticated encryption after the handshake are also derived from these unique values, ensuring that this one signature during the handshake effectively authenticates the entire session, no matter how much data is exchanged. Figure 11.4 shows the main messages exchanged in the TLS 1.3 handshake.



Figure 11.4 In the TLS handshake the server sends its own certificate and can ask the client for a certificate using a CertificateRequest message. The client responds with a Certificate message containing the certificate and a CertificateVerify message proving that it owns the associated private key.